

IBM HR Analytics Dashboard

Employee Attrition & Workforce Insights

Project Overview

This project analyzes employee attrition, workforce demographics, salary trends, job satisfaction, overtime impact, and business travel patterns using SQL and Power BI. The goal is to transform HR data into actionable business insights for workforce planning and retention strategies.

Business Objectives

- Analyze employee attrition patterns
- Identify departments and job roles with high attrition
- Study overtime impact on employee turnover
- Understand workforce demographics
- Examine job satisfaction and retention trends
- Build an interactive HR dashboard

Dataset & Technical Implementation

Dataset: IBM HR Analytics Employee Attrition Dataset

Records: 1,470 Employees

Key Fields: Age, Department, JobRole, Gender, MonthlyIncome, Attrition, BusinessTravel, OverTime, JobSatisfaction, MaritalStatus

Tools & Technologies

- MySQL
- Power BI
- DAX Measures
- Excel / CSV

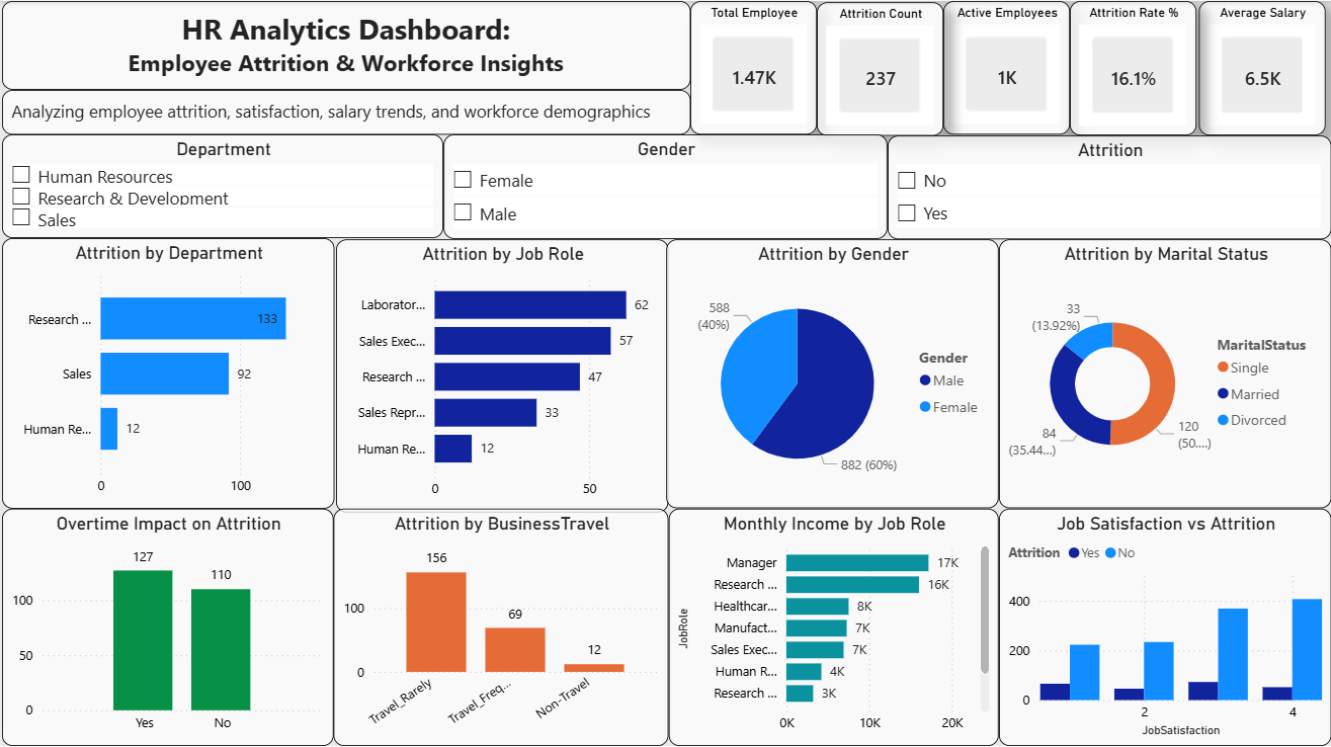
SQL Analysis Performed

- Row Count Validation
- NULL Value Analysis
- Duplicate Record Check
- Attrition Rate Calculation
- Attrition by Department & Job Role
- Overtime Impact Analysis
- Gender & Marital Status Analysis
- Salary Analysis
- Department Ranking using RANK() Window Function

KPIs Developed

- Total Employees
- Attrition Count
- Active Employees
- Attrition Rate (%)
- Average Salary

Power BI Dashboard



Key Insights

- Research & Development recorded the highest attrition.
- Employees working overtime showed higher turnover rates.
- Attrition was concentrated in specific job roles such as Laboratory Technicians and Sales Executives.
- Business travel patterns influenced employee retention.
- Workforce demographics and marital status revealed additional attrition trends.

Skills Demonstrated

SQL • Data Validation • Data Cleaning • Business Analysis • Power BI • DAX • Dashboard Development • Data Visualization • HR Analytics • Workforce Analytics

Project Outcome

This project demonstrates end-to-end Data Analytics skills by combining SQL-based analysis with interactive Power BI dashboards. The solution enables HR teams to monitor attrition trends, identify workforce risks, and support data-driven decision-making.